

PART 1 — PURE PYTHON MASTERY (40 QUESTIONS + LINKS)

A) BASIC PYTHON + PATTERN PROGRAMMING (10 QUESTIONS)

- 1. Print a square star pattern (5x5)

```
*****
*****
*****
*****
*****
```

- 2. Print a right-angle triangle pattern

```
*
**
***
****
*****
```

- 3. Print a pyramid pattern

```
  *
 ***
*****
*****
```

- 4. Print inverted pyramid pattern

```
*****
****
***
**
*
```

- 5. Print number triangle

```
1
12
123
1234
```

- 6. Find factorial of a number using loop

Link: <https://www.geeksforgeeks.org/python-program-for-factorial-of-a-number/>

- 7. Armstrong number check

Link: <https://www.geeksforgeeks.org/python-program-to-check-armstrong-number/>

- 8. Sum of digits of a number

Link: <https://www.geeksforgeeks.org/sum-of-digits-of-a-number/>

- 9. Print all prime numbers in a range

Link: <https://www.geeksforgeeks.org/python-program-to-print-all-prime-numbers-in-an-interval/>

- 10. Count vowels in a string

Link: <https://www.hackerrank.com/challenges/swap-case>

B) FUNCTIONS + RECURSION (8 QUESTIONS)

- 11. Fibonacci using recursion

Link: <https://www.geeksforgeeks.org/python-program-for-fibonacci-numbers/>

- 12. Reverse string using recursion
Link: <https://www.geeksforgeeks.org/reverse-string-using-recursion/>
- 13. Demonstrate *args and **kwargs
Link: <https://www.geeksforgeeks.org/args-kwargs-python/>
- 14. Create calculator using functions
- 15. Use lambda to sort list of tuples
Link: <https://www.geeksforgeeks.org/python-sort-tuples-list-by-second-value/>
- 16. Implement map/filter/reduce
Link: <https://www.geeksforgeeks.org/python-map-function/>
- 17. Count digits using recursion
- 18. Sum of list using recursion

C) PYTHON DATA STRUCTURES (7 QUESTIONS)

- 19. Remove duplicates from list
Link: <https://www.geeksforgeeks.org/python-ways-to-remove-duplicates-from-list/>
- 20. Find second largest in list
Link: <https://www.hackerrank.com/challenges/find-second-maximum-number-in-a-list>
- 21. Convert two lists into dictionary
Link: <https://www.geeksforgeeks.org/python-convert-two-lists-into-a-dictionary/>
- 22. Count frequency of words using dict
- 23. Demonstrate list comprehension to generate squares
- 24. Explain shallow vs deep copy
Link: <https://www.geeksforgeeks.org/copy-python-deep-copy-shallow-copy/>
- 25. Implement stack using list
Link: <https://www.geeksforgeeks.org/stack-in-python/>

D) STRING MANIPULATION + REGEX (5 QUESTIONS)

- 26. Reverse words in a string
Link: <https://leetcode.com/problems/reverse-words-in-a-string/>
- 27. Check if two strings are anagrams
Link: <https://leetcode.com/problems/valid-anagram/>
- 28. First non-repeating character
Link: <https://leetcode.com/problems/first-unique-character-in-a-string/>
- 29. Extract email IDs from text using regex
Link: <https://www.geeksforgeeks.org/python-check-email-address-valid-or-not/>
- 30. Count frequency of characters

E) PYTHON OOP QUESTIONS (5 QUESTIONS)

- 31. Create a BankAccount class
- 32. Demonstrate inheritance (Animal → Dog)
Link: <https://www.geeksforgeeks.org/inheritance-in-python/>
- 33. Demonstrate polymorphism
- 34. Use @staticmethod and @classmethod
Link: <https://www.geeksforgeeks.org/classmethod-in-python/>

- 35. Encapsulation with private variables

F) FILE HANDLING + EXCEPTION + MODULES (5 QUESTIONS)

- 36. Count lines in a file
- 37. Handle division by zero using try/except
- 38. Read JSON file and print keys
- 39. Create your own module and import it
- 40. Generate multiplication table in file

Why these questions?

These 40 Python questions build REAL mastery of the Python language — not DSA, not trick questions, but true understanding of syntax, functions, recursion, OOP, file handling, and patterns. Companies test whether you can THINK in Python, write clean logic, and understand fundamentals. Mastering these means you can tackle any coding round, explain confidently in interviews, and demonstrate complete command over Python — a must for 7–10 LPA backend or full■stack roles.